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# Remagnetizations and Thrusting in the Idaho-Wyoming Overthrust Belt

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**A study on the relationship between deformation and paleomagnetism was carried out in the frontal segment of the Idaho-Wyoming Overthrust Belt. Samples were taken from the Jurassic Twin Creek unit of the most easterly thrust, the Prospect Thrust, for paleomagnetic study. Detailed analysis of the paleomagnetic data showed a synfolding remagnetization with a paleopole (latitude/longitude: 83°N/286°E) located on the Tertiary segment of the North American apparent polar wander path (APWP). The age of the paleopole roughly coincides with the age of thrusting on the Prospect Thrust as determined by synorogenic deposits. The magnetization was found to reside in magnetite spherules. Previous data from the Absaroka and Darby Thrust sheets that lie to the west of the Prospect Thrust were reevaluated, and they also indicate a synfolding age for the remagnetization. The corresponding paleopole (64°N/162°E) lies on the late Mesozoic segment of the North American APWP. For these more interior thrust sheets, the apparent age of the remagnetization is older than that for the exterior Prospect Thrust sheet, but again roughly corresponds with the inferred age of thrusting for the sampled thrusts. We conclude that remagnetization in the Idaho-Wyoming Overthrust Belt is temporally related to deformation, and we suggest that authigenic magnetite growth occurred during discrete fluid pulses that were concurrent with thrusting and folding events.**

## INTRODUCTION

In recent years, a relationship between remagnetization and orogenic activity has become increasingly recognized. Oliver [1986] proposed that fluids, which were expelled from sediments by the action of overthrusting, may play a key role in such diverse phenomena as faulting, mineralization, migration of hydrocarbons, metamorphism, and remagnetization. Miller and Kent [1988] showed that there is a regional correlation between the timing of deformation and the timing of Permo-Carboniferous remagnetization of Appalachian limestone and redbed rock units, which they attributed to fluid migration patterns in space and time. In this paper we will document chronologically separate remagnetizations (late Mesozoic and Tertiary) in Jurassic rocks that occurred in different thrust sheets of the Idaho-Wyoming Overthrust Belt. This result shows that remagnetization is related to orogenic activity on scales that range from individual thrust sheets to major orogenic phases; in turn, this provides a potential tool to determine the timing of orogenic events [van der Pluijm et al., 1989].

The mechanism by which the Jurassic rocks were remagnetized can be inferred from the appearance of the magnetic carriers, which have been studied in extracts and were found to be spheroidal magnetites. These magnetite spherules are similar to those previously reported for the carbonates of the Appalachian foreland [e.g., McCabe et al., 1983; Bachtadse et al., 1987], which are thought to be of authigenic origin. While the link between spheroidal magnetites and fluid migration events remains speculative [McCabe and Elmore, 1989], alternative remagnetization mechanisms, such as thermal or thermoviscous resetting appear even less likely. Thus we tentatively suggest that the remagnetization events in the Idaho-Wyoming thrust belt were caused by discrete fluid migration pulses that are related to individual deformation events.

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## GEOLOGY AND TIMING OF THRUSTING

The Wyoming Salient is an arcuate section of the Idaho-Wyoming Overthrust Belt that deviates from the general N-S trend of the thrusts through most of Idaho and Wyoming in the area just south of Jackson, Wyoming where thrust traces strike NW-SE (Figure 1). This arcuate shape is considered to be the result of impingement of eastward moving thrust sheets on the crystalline-cored Teton and Gros Ventre uplifts [Dorr et al., 1977]. Deformation in the region has been primarily brittle with minor crystal plastic deformation (calcite twinning) and there is no evidence of metamorphism [Dorr et al., 1977; Wiltschko and Dorr, 1983].

The Idaho-Wyoming Overthrust Belt is one of the best dated orogenic systems. Wiltschko and Dorr [1983] have reviewed the sedimentological and paleontological evidence which constrains the timing of the major movements along the individual thrust planes of the belt. Starting in latest Jurassic-Early Cretaceous time with movements along the Paris Thrust in Idaho, a history of progressively younger movements to the east has been documented. For our study, the timing of the Absaroka, Darby, and Prospect thrust movements is of

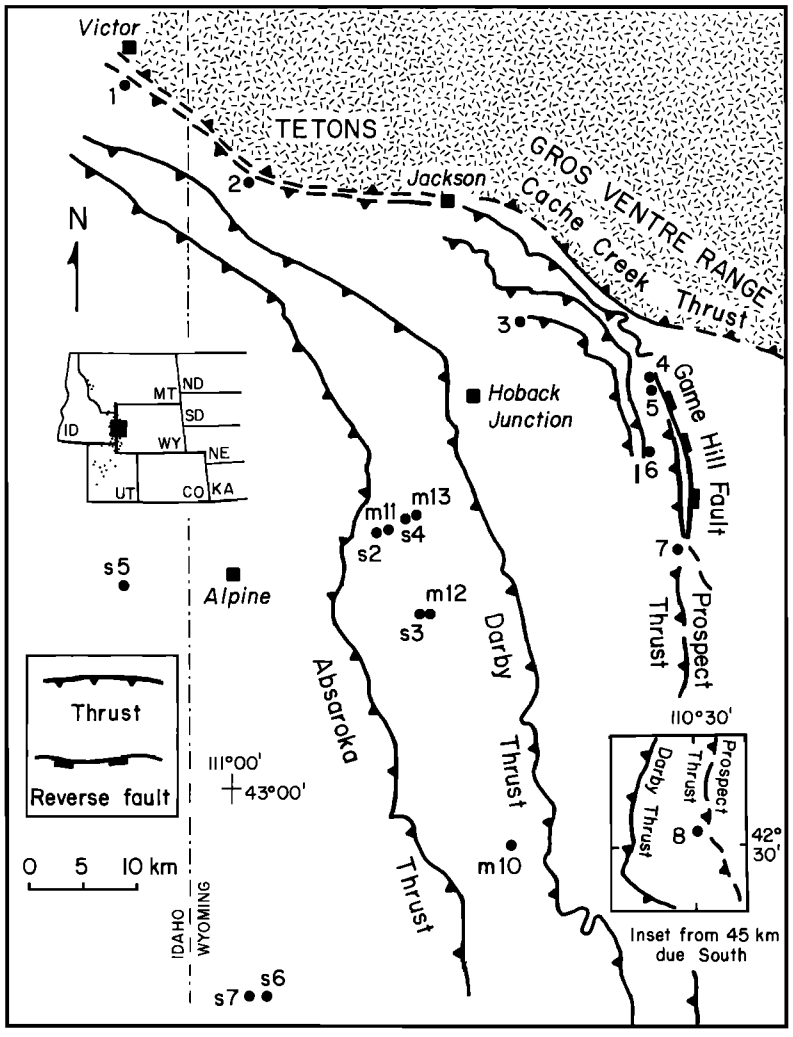



**Fig. 1. Sampling area in the Wyoming Salient of the Idaho-Wyoming Overthrust Belt showing the location of sites. Sites prefaced by S and M are from previous studies by Schwartz and Van der Voo [1984] and McCabe et al. [1982], respectively. Inset contains site 8 from a location 45 km due south of main sampling area.**

importance: major movements of the Absaroka Thrust sheet (Figure 1) occurred in the Late Cretaceous (Campanian-Maestrichtian) [Wiltschko and Dorr, 1983], whereas movements of the frontal Prospect Thrust is well constrained by synorogenic deposits to a much later time spanning the Paleocene-Eocene boundary [Dorr et al., 1977]. Kraig et al. [1987] proposed that major structures in the region now occupied by the intervening Darby Thrust sheet were formed by the westward movement of the unexposed Moxa basement thrust, at the same time as the translation of the Absaroka Thrust, although final movement on the Darby thrust may postdate even that of the Prospect. Previous paleomagnetic and calcite-twinning studies suggest large rotations in the Prospect sheet (up to 60° counterclockwise near the Teton Mountains and up to 30° clockwise near the Game Hill high-angle reverse fault in the east [Grubbs and Van der Voo, 1976; Craddock et al., 1988], while no significant rotation has been documented in the more interior, westerly thrust sheets [Grubbs and Van der Voo, 1976; Schwartz and Van der Voo, 1984; McCabe et al., 1982]. These authors have argued that buttressing effects by the uplifted foreland of the Teton-Gros Ventre-Game Hill Reverse Fault area caused the frontal thrusts to adjust to the buttress shape by rotating as observed, whereas the more interior thrusts did not experience such impediments to their lateral movements.

## PALEOMAGNETIC METHODS AND RESULTS

The 79 samples from eight sites along the frontal edge of the Prospect Thrust sheet were collected from the Jurassic Twin Creek Limestone unit by either drilling cores in the field or from oriented block samples that were drilled in the laboratory. The paleomagnetism of the Twin Creek Limestone had previously been studied by McCabe et al. [1982] in the Absaroka and Darby Thrust sheets. This study is an extension of the previous work and was designed specifically to study the effects of deformation in the more easterly Prospect Thrust sheet upon the remanence. The Twin Creek Limestone contains abundant marine fossils which indicate a Middle Jurassic age [Imlay, 1952]. The

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limestone is mainly composed of calcareous shales with intercalated grey to bluish micrites and oolites [Wright, 1971]. Our eight sampling sites are located along the margin of the Wyoming Salient in the Prospect Thrust sheet, from just south of Victor, Idaho, to west of Big Piney, Wyoming, at 42.5°N (Figure 1).

Samples were subjected to stepwise thermal demagnetization using a Schonstedt TSD-1 furnace or alternating field (AF) demagnetization with a Schonstedt GSD-1 demagnetizer. Measurements of the remanence were made with a ScT superconducting magnetometer. Demagnetization results were plotted in orthogonal diagrams for vector analysis using the techniques outlined by Kirschvink [1980].

Demagnetization of the Twin Creek samples generally revealed three components of magnetization (Figure 2). The first component is usually removed at temperatures below 150°C and in fields below 10 mT.

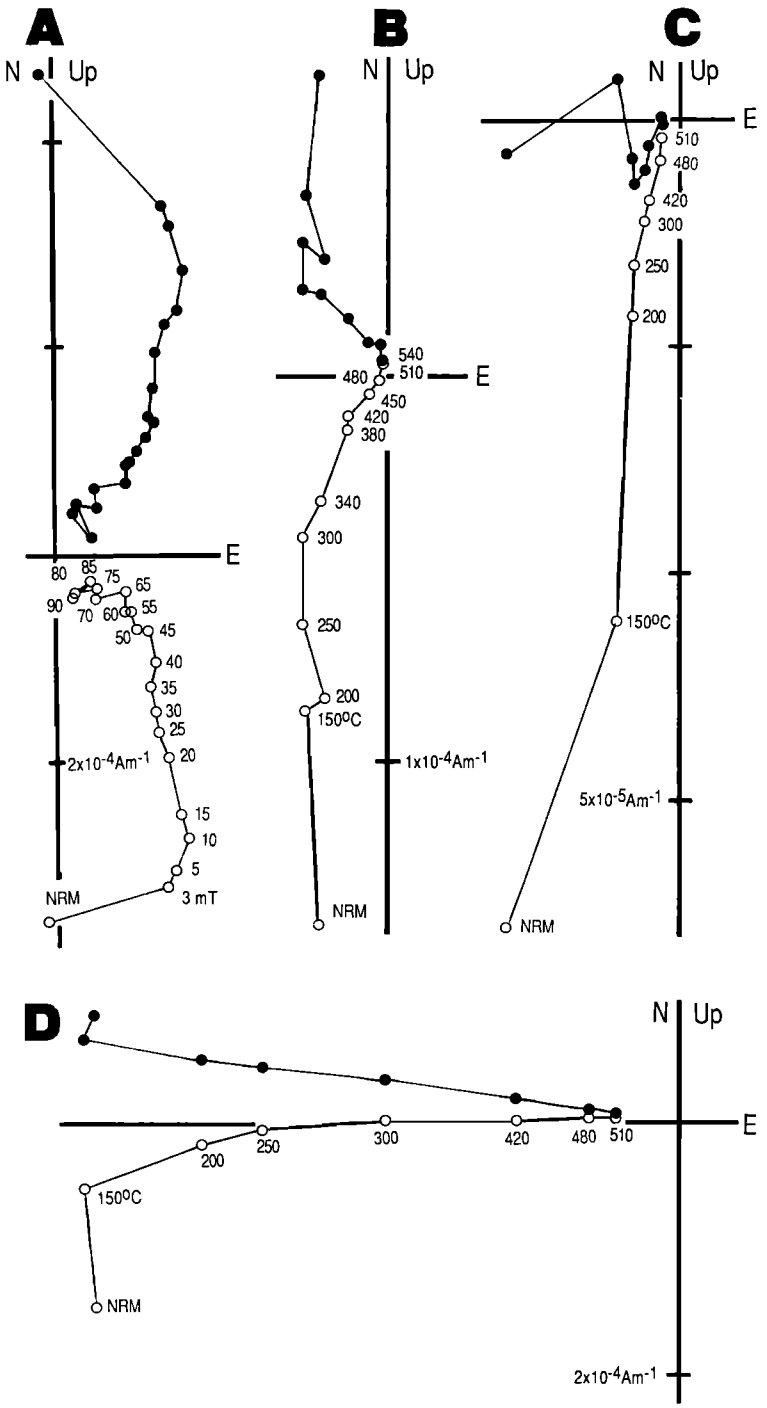



**Fig. 2. Orthogonal demagnetization diagrams showing the endpoints of the magnetization vectors. (a) Sample from site 1 was subjected to stepwise alternating field (AF) demagnetization. (b), (c), (d) Samples from sites 7, 6 and 4 were demagnetized by stepwise thermal demagnetization. All directions are plotted in in situ coordinates. Solid circles represent projections on the horizontal plane and open circles represent those on the east-west vertical plane.**

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The orientations of this first component of magnetization are highly scattered in in situ coordinates and are interpreted to be a soft viscous magnetization acquired in the laboratory. Once the soft magnetization is removed, the second magnetization is eliminated in alternating fields between 10 and 45 mT and at temperatures between 150° and 300°C (e.g., Figure 2). This component is usually northerly and down in in situ coordinates and conforms to the present-day geomagnetic field direction in Wyoming. We interpret this component to be a viscous magnetization of recent origin. The third component is defined by univectorial decay to the origin of the vector diagram and is the characteristic magnetization for the unit. Thermal demagnetization of this characteristic component was continued until the intensity of magnetization had decreased to below the noise level; generally by 480°-540°C no appreciable magnetization remains in the samples. All of the characteristic directions are of normal polarity; the site means are given in Table 1. The 19 samples were not used to calculate site mean directions as they yielded noisy data and uninterpretable demagnetization diagrams.

The magnetic carrier in the Twin Creek limestone is magnetite, in agreement with the previous conclusion of McCabe et al. [1982]. McCabe and co-workers also found evidence for goethite (in at least one site) and minor hematite, although the latter did not appear to carry any remanence as indicated by the general absence of high coercivities and blocking temperatures. A magnetic extract obtained by the technique described by McCabe et al. [1983] revealed magnetite spherules in our samples ranging in size from 10 to 100 micrometer (Figure 3); we note that smaller (single-domain sized) grains may remain undetected by this extraction technique. While not specifically tested in our study, such magnetite spherules have previously been shown to be polycrystalline aggregates, suggesting that portions of these spherules may well be of single-domain nature. Low-temperature transitions, indicative of (multidomain) magnetocrystalline anisotropy, are absent in Twin Creek samples, further suggesting that the magnetite carrier may be single domain [McCabe et al., 1982]. Energy-dispersive analysis of the magnetite spherules reveals no cations other than iron, indicating that titanium is not present in significant quantities.

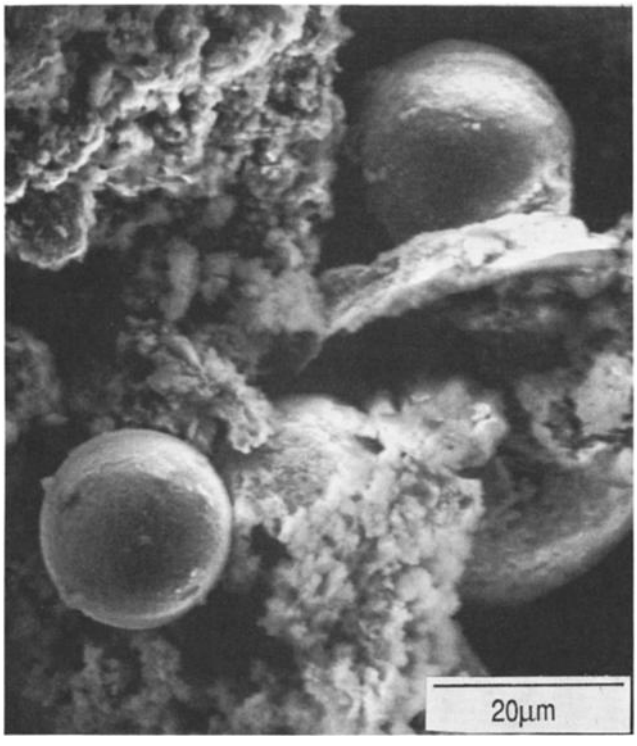



**Fig. 3. Scanning electron micrograph showing magnetite spherules extracted from a sample of Twin Creek Limestone at site 4.**

Characteristic directions for the samples from the Prospect Thrust show a high degree of scatter when plotted in in situ coordinates, but when tilt correction is applied clustering improves significantly (Figure 4). In a stepwise foldtest, clustering reaches a peak at 80% tilt correction and then decreases again as tilt correction continues to the 100% level (Figure 4).

Site means from data collected by Schwartz and Van der Voo [1984] in the Jurassic Stump unit and by Hudson et al. [1989] in the Jurassic Preuss Sandstone also reveal such behavior. We have reanalyzed the data of McCabe et al. [1982] from the Jurassic Twin Creek of the Darby and Absaroka thrusts using an incremental foldtest. When this previous study was being prepared for publication, remagnetization of carbonates had not yet been widely recognized and incremental fold tests [e.g., Schwartz and Van der Voo, 1984] were not yet routinely applied. Even though the previous Twin Creek magnetic directions pass the (100%) foldtest at the 99% confidence level, our reexamination of the data suggested that the directions bypass each other during tilt correction. The site means for all three studies are given in Tables 1 and 2 and are shown in Figure 5. The incremental foldtests are illustrated in Figure 6; site locations are shown in Figure 1. As can be seen in Figure 5, site means cluster poorly before tectonic tilt correction (circles) and appear to bypass each other on the way to full tilt correction (squares). Plotting k versus percent unfolding (Figure 6), an apparently synfolding peak at about 80% can be observed.

During the fold test [McElhinny, 1964] the Fisher precision parameter (k) for the Twin Creek site means from the Prospect Thrust shows a peak value at about 80% unfolding which is statistically significant at the 90% confidence level (Figure 6); this statistical significance is with respect to the k value of 20 at full (100%) unfolding, whereas with respect to the k value of 3 before any structural correction is applied (0% unfolding), the test is significant at the 99% confidence level (Table 1). The best grouped mean direction is (D/I) 6°/+66° ($\alpha_{95}$=9, k=42). For the reanalyzed data of the Absaroka and Darby thrusts the best grouped mean direction is (D/I) 325°/+58°, ($\alpha_{95}$=8, k=43). For the latter results the peak value of k at 80% unfolding is significant at the 75% confidence level, again with respect to the k value of 29 at 100% unfolding. As before, the maximum k value of 43 at 80% unfolding is significantly different at the 99% confidence level from the k value of 4 without any structural correction (Table 2). For both the Prospect and Absaroka/Darby Twin Creek results, we can reject the hypothesis that the magnetizations were acquired after the folding, i.e., the need for a

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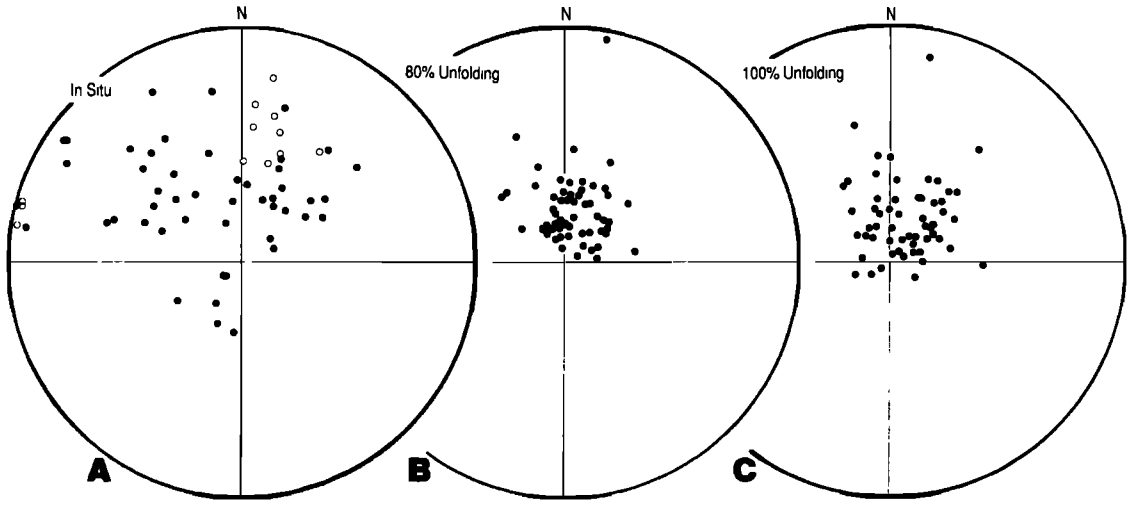



**Fig. 4. Equal-angle projections of the characteristic directions for 60 samples taken from the Twin Creek unit in the Prospect Thrust sheet. Directions are plotted with (a) 0%, (b) 80%, and (c) 100% tilt correction. Open circles represent upward oriented (negative) vectors; solid circles represent downward oriented (positive) vectors.**

structural correction is clearly indicated by the incremental fold test. On the other hand, we cannot conclude that the clustering of directions at 80% and 100% unfolding is statistically significantly different at the customary 95% confidence level; instead, the significance is reduced to the 90% level for the Prospect and 75% for the Absaroka/Darby results. While we can conclude, therefore, that the magnetization was acquired before the folding was complete, we cannot statistically prove that the magnetization was acquired during the folding. However, this may be a problem whenever a peak in k value lies close to 100% unfolding and does not preclude a synfolding conclusion [van der Pluijm, 1987]. Moreover, as we will show below, the directions of magnetization (and the paleopole) of the Prospect Thrust suggest conclusively that the age of magnetization is significantly later than Jurassic and that the magnetization is therefore secondary.

## DISCUSSION AND CONCLUSIONS

In a paleomagnetic study on the Glarus Thrust of the Swiss Alps, internal deformation associated with thrust sheet translation was shown to produce a distinct

TABLE 1. Jurassic Prospect Site Means

| Site | N/No | In Situ | Tilt-Corrected | | Bedding | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|
| | | | 80% | 100% | | | |
| | | D/I | D/I | D/I | DD/D | | |
| 1 | 13/16 | 35/+42 | 18/+64 | 7/+69 | 237/32 | 12 | 12 |
| 2 | 8/8 | 14/-34 | 8/+65 | 280/+85 | 020/35* | 44 | 8 |
| 3 | 7/12 | 21/+66 | 27/+72 | 28/+73 | 182/07 | 30 | 11 |
| 4 | 9/12 | 292/+ 4 | 5/+67 | 43/+57 | 269/70* | 29 | 10 |
| 5 | 6/9 | 311/+48 | 24/+62 | 41/+55 | 91/58 | 60 | 9 |
| 6 | 6/6 | 214/+67 | 351/+73 | 359/+64 | 016/46 | 33 | 12 |
| 7 | 6/8 | 337/+29 | 26/+67 | 61/+68 | 133/65 | 19 | 16 |
| 8 | 5/8 | 296/+34 | 332/+45 | 349/+43 | 058/52 | 48 | 11 |

| Unfolding, % | Overall Mean D/I | k | $\alpha_{95}$ | Paleopole Position at 80% Tilt Correction (latitude/longitude): 83°/286° |
|---|---|---|---|---|
| 0 | 308/+59 | 3 | 38 | |
| 80 | 6/+66 | 42 | 9 | dp=12° dm=15° |
| 100 | 18/+68 | 20 | 13 | |

Jurassic Twin Creek site means. N/No is the number of sample directions used in the calculation of the site means versus the number of samples demagnetized. D/I are declination and inclination, DD/D are dip direction and dip, k and $\alpha_{95}$ are statistical parameters [Fisher, 1953]. Polar error signified by dp and dm demarks the oval of 95% confidence about the pole position [McElhinny, 1973].

* Overturned beds.

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TABLE 2. Jurassic Absaroka/Darby Site Means

| Site | N/No | In Situ | Tilt-Corrected | | Bedding | k | $\alpha_{95}$ |
|---|---|---|---|---|---|---|---|
| | | | 80% | 100% | | | |
| | | D/I | D/I | D/I | DD/D | | |
| S2 | 4/7 | 270/+15 | 314/+60 | 343/+60 | 060/80 | - | - |
| S3 | 8/8 | 354/+43 | 334/+47 | 329/+47 | 244/25 | 16 | 14 |
| S4 | 9/9 | 10/+47 | 325/+59 | 310/+57 | 238/43 | 105 | 5 |
| S5 | 6/8 | 334/+40 | 308/+59 | 298/+61 | 191/35 | 22 | 13 |
| S6 | 8/9 | 32/+49 | 345/+73 | 318/+73 | 237/41 | 168 | 4 |
| S7 | 9/9 | 276/+31 | 339/+67 | 7/+63 | 069/70 | 32 | 9 |
| M10 | 8/11 | 17/+48 | 343/+54 | 335/+54 | 258/34 | 22 | 12 |
| M11 | 8/11 | 285/-23 | 304/+48 | 332/+55 | 248/66* | 49 | 8 |
| M12 | 5/11 | 292/+32 | 341/+45 | 353/+40 | 058/70 | 70 | 9 |
| M13 | 6/6 | 10/+60 | 310/+61 | 297/+57 | 249/40 | 34 | 13 |

| Unfolding, % | Overall Mean D/I | k | $\alpha_{95}$ | Paleopole Position at 80% Tilt Correction (latitude/longitude): 64°/162° |
|---|---|---|---|---|
| 0 | 324/+44 | 4 | 29 | |
| 80 | 325/+58 | 43 | 8 | dp=8° dm=11° |
| 100 | 329/+59 | 29 | 9 | |

Jurassic site means from previous studies used in this paper for the Absaroka/Darby thrust sheet. Sites prefaced with S and M are from Schwartz and Van der Voo [1984] and McCabe et al. [1982], respectively. N/No is the number of sample directions used in the calculation of the site means versus the number of samples demagnetized. D/I are declination and inclination, DD/D are dip direction and dip, $\alpha_{95}$ and k are statistical parameters [Fisher, 1953]. Polar error signified by dp and dm demarks the oval of 95% confidence about the pole position [McElhinny, 1973].

* Overturned beds.

great circle pattern in stereographic projection, which contains the magnetic direction predicted from the paleopole and the observed tilt-corrected magnetic directions [Hirt et al., 1986]. Such behavior can be approximated by the trajectory of a passive line during progressive simple shear [e.g., Hobbs et al., 1976, p. 192]. The magnetic vector will rotate to or away from the thrusting direction dependent on their angular relationship. In the case of a Jurassic magnetization in the Idaho-Wyoming Overthrust Belt, the vector means would deflect along a great circle path in a westerly and shallowing direction in response to east-directed thrusting (Figure 7). However, the Twin Creek mean for the Prospect sheet lies in the opposite direction to that predicted for such an internal deformation mechanism. Furthermore, we do not find structural evidence for grain rotation or plastic deformation in the Twin Creek other than low strain twinning [Craddock et al., 1988], and brittle deformation has been the primary mode of stress accommodation in the Wyoming Salient [Wiltsch-

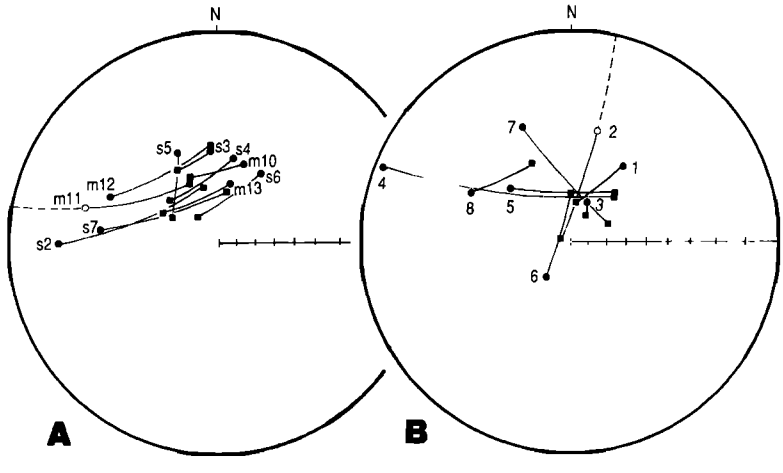



**Fig. 5. Equal-angle projections of the site means for (a) Absaroka/Darby and (b) Prospect sheets. Site means prefaced by S and M are from previous studies by Schwartz and Van der Voo [1984] and McCabe et al. [1982], respectively. A summary of site mean directions is contained in Table 2. Circles are in situ site mean directions which correct to the squares when 100% tilt correction is applied.**

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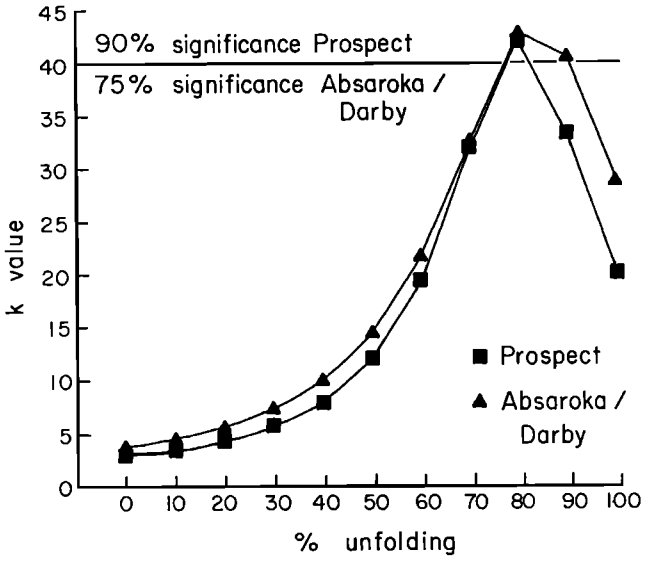



Fig. 6. Incremental fold tests [McElhinny, 1964] for sites in the Prospect and the Absaroka/Darby sheets; the Fisher precision parameter (k) is plotted versus the percent unfolding. The 90% significance level for the $k_{max}$ of Prospect sites and 75% significance level for Absaroka/Darby sites were calculated with respect to the k values at 100% unfolding.

ko and Dorr, 1983]. Thus we argue that internal deformation cannot be held responsible for the apparent synfolding nature of the magnetization.

The timing of magnetization acquisition in the Prospect sheet is constrained by stuctural as well as paleomagnetic evidence. As argued earlier, the notion that the Prospect Thrust magnetization in the Twin Creek was acquired after the folding can be rejected. We can therefore bracket the age of the Twin Creek magnetization to the time interval between deposition in the Middle Jurassic and the Eocene time of folding. The pole position determined from the Prospect Thrust directions, either at 100% or 80% unfolding, falls on the North American Apparent Polar Wander Path segment for the Tertiary to Recent [Harrison and Lindh, 1982] and is significantly different from the Jurassic or Cretaceous paleopoles (Figures 7 and 8). Combined with the above arguments, the age of the magnetization is then constrained to be post-Cretaceous to pre-Oligocene. If we accept a synfolding age of magnetization, acquired early in the deformation history (because of the peak at 80% unfolding), we would argue that the onset of folding coincided with the remagnetization event recorded in the Twin Creek of the Prospect Thrust. An Early Tertiary age for this event is then most likely.

Problematic with this interpretation remains the absence of significant rotations recorded in the Twin Creek. Grubbs and Van der Voo [1976] published results for Triassic units in the study area which showed that the Prospect sheet rotated at least 60° counterclockwise in the northwestern region, north of Hoback Junction, and 30° clockwise in the area southeast of Hoback Junction. The Triassic red beds studied by Grubbs and Van der Voo [1976] are clearly not remagnetized in the Paleocene. The directions of the Triassic Ankareh Formation are of dual polarity with inclinations uniformly around ca. 20°, which are very different from the directions observed in the Twin Creek Formation. We speculate that hematite in the Ankareh red

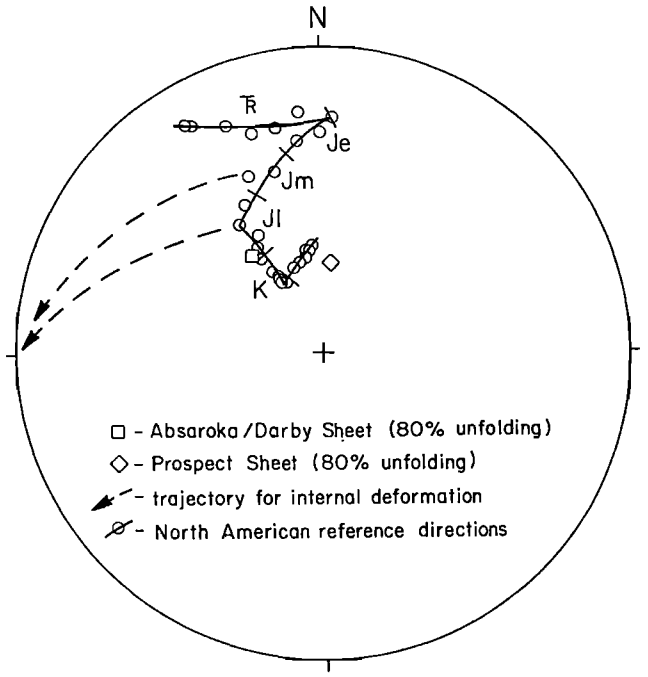



Fig. 7. Equal-area projection showing the expected great circle trajectory for middle and late Jurassic directions if eastward thrusting was accompanied by internal deformation in the thrust sheet. The North American reference directions for Triassic to Tertiary times were calculated for the Wyoming locality from the data given by Harrison and Lindh [1982] and May and Butler [1986]. Solid square and diamond represent the Twin Creek magnetic directions for the Absaroka/Darby and Prospect Thrusts at 80% tilt correction, respectively.

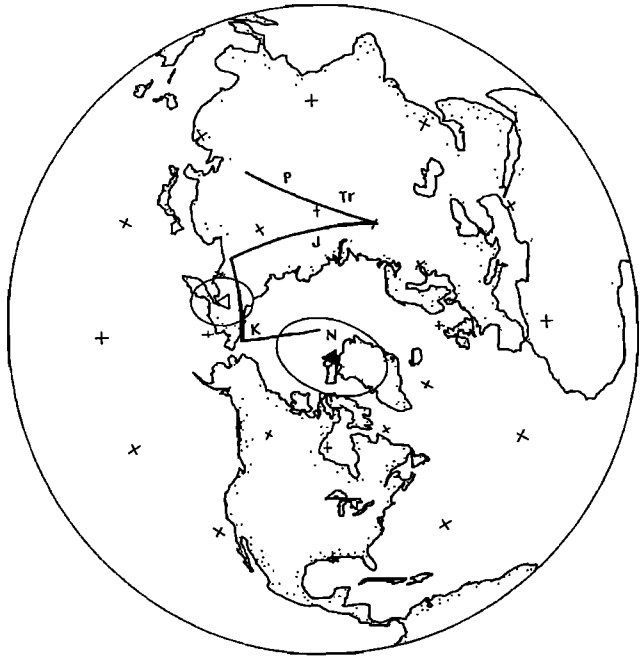



Fig. 8. Pole positions of Jurassic Twin Creek sites in the Prospect sheet (solid triangle; located in North Greenland) and Absaroka/Darby sheet (open triangle; located near Kamchatka Peninsula) that were calculated from the means at 80% unfolding. Solid line represents the North American apparent polar wander path (APWP) compiled from May and Butler [1986] and Harrison and Lindh [1982].

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beds formed early and that this mineral remained stable, both mineralogically and magnetically, under the conditions that caused remagnetization of the carbonates. The directions in the Twin Creek Limestone, however, do not show the rotations observed in the Triassic red bed directions; if they did, our northern sites (1-3) would have more westerly declinations than the southern sites (4-8), but inspection of (80% and 100%) tilt-corrected directions in Table 1 shows that that is not the case. Thus we have the following paradox: prefolding directions of the Triassic show thrust sheet rotations in the Prospect, but directions from the synfolding remagnetized Twin Creek Formation do not show rotations; yet one would assume that rotations and folding all occurred at the time of thrusting. We have no solution for this paradox with the currently available data. A conclusion that rotation occurred before folding of the Prospect Thrust sheet is one possibility. Another possible conclusion could be that the deviating declinations in the Triassic rocks do not record thrust sheet rotations, but rather reflect the effect of strain on the directions. In future studies of rocks in this area, we hope to devise tests for these ideas.

The fold test for the Jurassic sites in the Absaroka/Darby sheet also indicates an early synfolding remagnetization (Figure 6). In contrast to the result from the Prospect Thrust, the pole position for the mean of the Absaroka/Darby site means, at either 80% or 100% unfolding (Table 2), falls on the late Mesozoic segment of the North American Apparent Polar Wander Path of May and Butler [1986], as is shown in Figures 7 and 8. Given the location of the paleopole, one could argue that the Twin Creek of the Absaroka/Darby sheet is not remagnetized at all, with the minor crossover observed during structural correction being the result of small, postfolding present-day field components that contaminate the characteristic directions of Figure 5a and Table 2. We have two arguments against this conclusion. First, Schwartz and Van der Voo [1984] and Hudson et al. [1989] also obtained results from Jurassic and Cretaceous rocks of the Absaroka/Darby sheet that give a statistically significant syn-folding magnetization of approximately Late Mesozoic age. Second, a small but unrecognized contamination by a present-day field overprint in the characteristic directions cannot be a suitable explanation for all directions: depending on the geometry of the bedding attitudes, the effect of such a hypothetical contamination is to bring the characteristic direction closer to that of the present-day field along a greatcircle trajectory. For many of the site means shown in Figure 5b, this is a plausible explanation for the crossover (the apparently synfolding distribution), but for sites 8, 9, and 14 of the Cretaceous rocks studied by Schwartz and Van der Voo [1984] and sites S3, S5, S6, and M13 of Figure 5a and Table 2, correction for a hypothetical present-day field overprint would remove the tilt-corrected directions from the common cluster of the site means and would not lead to an improvement in the precision parameter k. Finally, the fact remains that all directions in the Middle to Upper Jurassic rocks of the Absaroka/Darby thrust sheets are of normal polarity, whereas many reversals have been documented for this time [e.g., Steiner et al., 1985/86, 1987], and that the paleopole obtained (Table 2) falls on the latest Jurassic (?) to Cretaceous segment of the apparent polar wander path, younger than one would expect for the age of the rocks (Figures 7 and 8). A Cretaceous age correlates with the timing of major movement on the Absaroka Thrust [Wiltschko and Dorr, 1983] and the contemporaneous deformation of the Darby due to the effects of the westward moving Moxa Thrust, as proposed by Kraig et al. [1987]. We thus conclude that the different mean directions and paleopoles for the Prospect and Darby/Absaroka thrusts were caused by magnetization acquisition at different times, i.e., the Early Tertiary and the Late Mesozoic, respectively.

The two secondary magnetizations acquired by the Jurassic units can be either of thermoviscous or chemical origin. Burial temperatures for the thrust sheets ranged from 125°-200°C [Edman and Surdam, 1984]. Assuming that the maximum temperature of 200°C lasted for a period of $2 \times 10^8$ years, then laboratory blocking temperatures of about 450°C would be expected [Middleton and Schmidt, 1982], which is 60° lower than the blocking temperatures we observed in the laboratory (510°C). This would suggest that the magnetization observed in the thrust sheets is unlikely to have been of thermo-viscous origin. However, Kent [1985] has argued that there is a large degree of uncertainty about the time/temperature relationship for thermoviscous remagnetization of magnetites in limestones. Thus thermoviscous magnetization cannot be entirely ruled out for the origin of the observed synfolding remagnetizations. Yet, since we attribute the magnetization of the Jurassic units to the presence of authigenic magnetites, including the spherules shown in Figure 3, which are clearly secondary in origin [Benthien and Elmore, 1987; Elmore et al., 1987; McCabe et al., 1983, 1987], we feel that chemical remagnetization is the most likely interpretation. Several chemical mechanisms for the formation of magnetite spherules have been proposed [McCabe et al., 1987], but with the currently available observations of Twin Creek magnetites, we are unable to choose between such causes as biodegradation of hydrocarbons, release of iron from carbonate or layer silicate minerals, oxidation of framboidal pyrite, or reduction of precursor oxides. What is clear, however, is that each of these mechanisms of magnetite formation requires the interaction of a fluid with the rock unit. In future studies we will include isotopic and electron microscopy techniques in order to obtain further evidence for the nature of such fluids and how they interacted with the carbonates of the Twin Creek Formation in the Absaroka/Darby and Prospect thrust sheets at different times.

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